Chapter 02. Capital Allocation to Risky Assets

Shushu Liao

Investment I

Peking University Business School UK Campus

Chapter Outline

- Stock Risk and Return
- Risk aversion and its estimation
- Capital allocation between risky and risk-free assets
- Passive strategies and the capital market line (CML)

Holding Period Return

- Rates of return: Single period

$$\text{HPR} = \frac{P_1 - P_0 + D_1}{P_0}$$

- HPR = Holding period return (In practice, we use adjusted close price in the following period (adjusted for distribution) as the proxy for $D_1 + P_1$)
- P_0 = Beginning price
- P_1 = Ending price
- D_1 = Dividend during period one

Time Series Analysis of Past Rates of Return

- True means and variances are unobservable because we don't actually know possible scenarios like the one in the examples
- So we must estimate them (the means and variances, not the scenarios)

Returns Using Arithmetic and Geometric Averaging

- Arithmetic Average

$$E(r) = \frac{1}{n}\sum_{s=1}^{n} r(s)$$

- Geometric Average

$$TV = (1 + r_1)(1 + r_2)\ldots(1 + r_n)$$

$$g = (TV)^{\frac{1}{n}} - 1$$

Estimating Variance and Standard Deviation

- Estimated Variance
  - Expected value of squared deviations

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{s=1}^{n}[r(s) - \bar{r}]^2$$

- Unbiased estimated standard deviation

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{j=1}^{n}[r(s) - \bar{r}]^2}$$

Annualized Return and Volatility

- To make it more comparable, in general, we annualize the returns
- Annualized Mean and Standard Deviation

Annualized Standard Deviation = Standard Deviation of Daily Returns x $\sqrt{252}$

Annualized Standard Deviation = Standard Deviation of Monthly Returns x $\sqrt{12}$

Annualized Variance/Covariance = Variance/Covariance of Daily Returns x 252

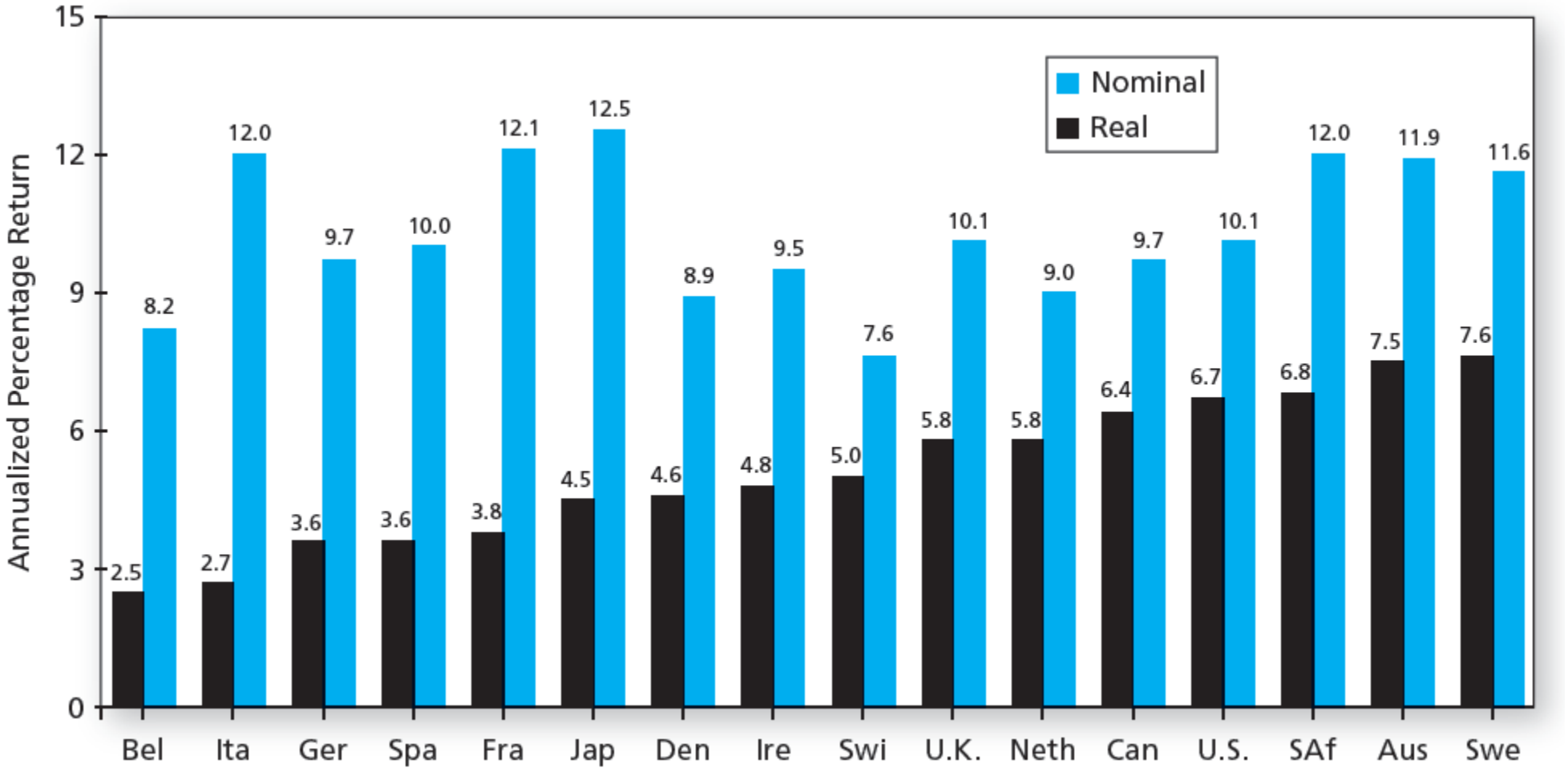
Annualized Variance/Covariance = Variance/Covariance of Monthly Returns x 12

Annualized Return=(1+ Average Daily returns)^252-1 ≈ Average Daily returns x 252

Annualized Return =(1+ Average Monthly returns)^12-1 ≈ Average Monthly returns x 12

*there are 252 trading days in a year

Nominal and Real Equity Returns Around the World, 1900-2000



The Reward-to-Volatility (Sharpe) Ratio

- Excess Return
  - The difference in any particular period between the actual rate of return on a risky asset and the actual risk-free rate
- Risk Premium
  - The difference between the expected HPR on a risky asset and the risk-free rate
- Sharpe Ratio

$$\frac{\text{Risk premium}}{\text{SD of excess returns}}$$

Risk and Risk Aversion

- Utility Values
  - Investors are willing to consider:
    - Risk-free assets
    - Speculative positions with positive risk premiums
  - Portfolio attractiveness increases with expected return and decreases with risk
  - What happens when return increases with risk?

Available Risky Portfolios

- How to measure which portfolio is better: Each portfolio receives a utility score to assess the investor's risk/return trade off

Table 6.1

Available risky portfolios (Risk-free rate = 5%)

Portfolio	Risk Premium	Expected Return	Risk (SD)
L (low risk)	2%	7%	5%
M (medium risk)	4	9	10
H (high risk)	8	13	20

Utility Function

- Utility Function
  - U = Utility
  - $E(r)$ = Expected return on the asset or portfolio
  - A = Coefficient of risk aversion
  - σ^2 = Variance of returns
  - ½ = A scaling factor

$$U = E(r) - \tfrac{1}{2} A\sigma^2$$

The Investor's Risk Tolerance

- What are reasonable values for A?

  This is a tricky question and the subject of much debate among financial economists

  Financial investment advisors often rely on questionnaires to deduce individual's risk tolerance and the value of A

The Investor's Risk Tolerance

Asset Allocation Planner Step 1

Your Personal Risk Tolerance

Your prior investment experience can help determine your attitude toward investment risk.

10 Have you ever invested in individual bonds or bond mutual funds? (Aside from U.S. savings bonds.)

- ☐ No, and I would be uncomfortable with the risk if I did.
- ☐ No, but I would be comfortable with the risk if I did.
- ☐ Yes, but I was uncomfortable with the risk.
- ☐ Yes, and I felt comfortable with the risk.

11 Have you ever invested in individual stocks or stock mutual funds?

- ☐ No, and I would be uncomfortable with the risk if I did.
- ☐ No, but I would be comfortable with the risk if I did.
- ☐ Yes, but I was uncomfortable with the risk.
- ☐ Yes, and I felt comfortable with the risk.

Your comfort level with investment risk influences how aggressively or conservatively you choose to invest. It should be balanced with the potential of achieving your investment goals.

12 Which ONE of the following statements best describes your feelings about investment risk?

- ☐ I would only select investments that have a low degree of risk associated with them (i.e., it is unlikely I will lose my original investment).
- ☐ I prefer to select a mix of investments with emphasis on those with a low degree of risk and a small portion in others that have a higher degree of risk that may yield greater returns.
- ☐ I prefer to select a balanced mix of investments -- some that have a low degree of risk, others that have a higher degree of risk that may yield greater returns.
- ☐ I prefer to select an aggressive mix of investments -- some that have a low degree of risk, but with emphasis on others that have a higher degree of risk that may yield greater returns.
- ☐ I would select an investment that has only a higher degree of risk and a greater potential for higher returns.

13 If you could increase your chances of improving your returns by taking more risk, would you:

- ☐ Be willing to take *a lot* more risk with *all* your money.
- ☐ Be willing to take *a lot* more risk with *some* of your money.
- ☐ Be willing to take *a little* more risk with *all* your money.
- ☐ Be willing to take *a little* more risk with *some* of your money.
- ☐ Be unlikely to take much more risk.

Back to the Table

- For portfolio L, M and H, we compute the utility scores.
- The choice of portfolio depends on investor's risk aversion.

Investor Risk Aversion (A)	Utility Score of Portfolio L [$E(r) = .07$; $\sigma = .05$]	Utility Score of Portfolio M [$E(r) = .09$; $\sigma = .10$]	Utility Score of Portfolio H [$E(r) = .13$; $\sigma = .20$]
2.0	$.07 - ½ \times 2 \times .05^2 = .0675$	$.09 - ½ \times 2 \times .1^2 = .0800$	$\mathbf{.13 - ½ \times 2 \times .2^2 = .09}$
3.5	$.07 - ½ \times 3.5 \times .05^2 = .0656$	$\mathbf{.09 - ½ \times 3.5 \times .1^2 = .0725}$	$.13 - ½ \times 3.5 \times .2^2 = .06$
5.0	$.07 - ½ \times 5 \times .05^2 = .0638$	$\mathbf{.09 - ½ \times 5 \times .1^2 = .0650}$	$.13 - ½ \times 5 \times .2^2 = .03$

Estimating Risk Aversion

- Mean-Variance (M-V) Criterion
  - Portfolio A dominates portfolio B if:

$$E(r_A) \geq E(r_B)$$

and

$$\sigma_A \leq \sigma_B$$

Capital Allocation Across Risky and Risk-Free Portfolios

- Asset Allocation
  - The choice among broad asset classes that represents a very important part of portfolio construction
- The simplest way to control risk is to manipulate the fraction of the portfolio invested in **risk-free assets** versus the portion invested in the **risky assets**

Basic Asset Allocation Example

Risk-free money market fund	$90,000
Equities	$113,400
Bonds (long-term)	$96,600
Total risk assets	$210,000
Total market value	$300,000

Risk assets

- Denote w as the weight on equities, and $(1 - w)$ is the weight on bond
- $w = \frac{113{,}400}{210{,}000} = 0.54$
- $1 - w = 0.46$

Basic Asset Allocation Example

y = Weight of the risky portfolio, P, in the complete portfolio

$1 - y$ = Weight of risk-free assets

$$y = \frac{\$210{,}000}{\$300{,}000} = 0.7 \qquad 1 - y = \frac{\$90{,}000}{\$300{,}000} = 0.3$$

The Risk-Free Asset

- Only the government can issue default-free securities
- T-bills viewed as “the” risk-free asset
- Money market funds also considered risk-free in practice

Portfolios of One Risky Asset and a Risk-Free Asset

- It's possible to create a complete portfolio by splitting investment funds between safe and risky assets
- Let
  - y = Portion allocated to the risky portfolio, P
  - (1 - y) = Portion to be invested in risk-free asset, F
- Complete portfolio return is:
  - $r_C = yr_p + (1 - y)r_f$

One Risky Asset and a Risk-Free Asset: Example

$\mathrm{r_f} = 7\%$ $\sigma_{rf} = 0\%$

$\mathrm{E(r_p)} = 15\%$ $\sigma_p = 22\%$

The expected return on the complete portfolio:

$$E(r_c) = 15y + 7(1 - y) = 7 + y(15 - 7)$$

The risk of the complete portfolio:

$$\sigma_C = y\sigma_P = 22y$$

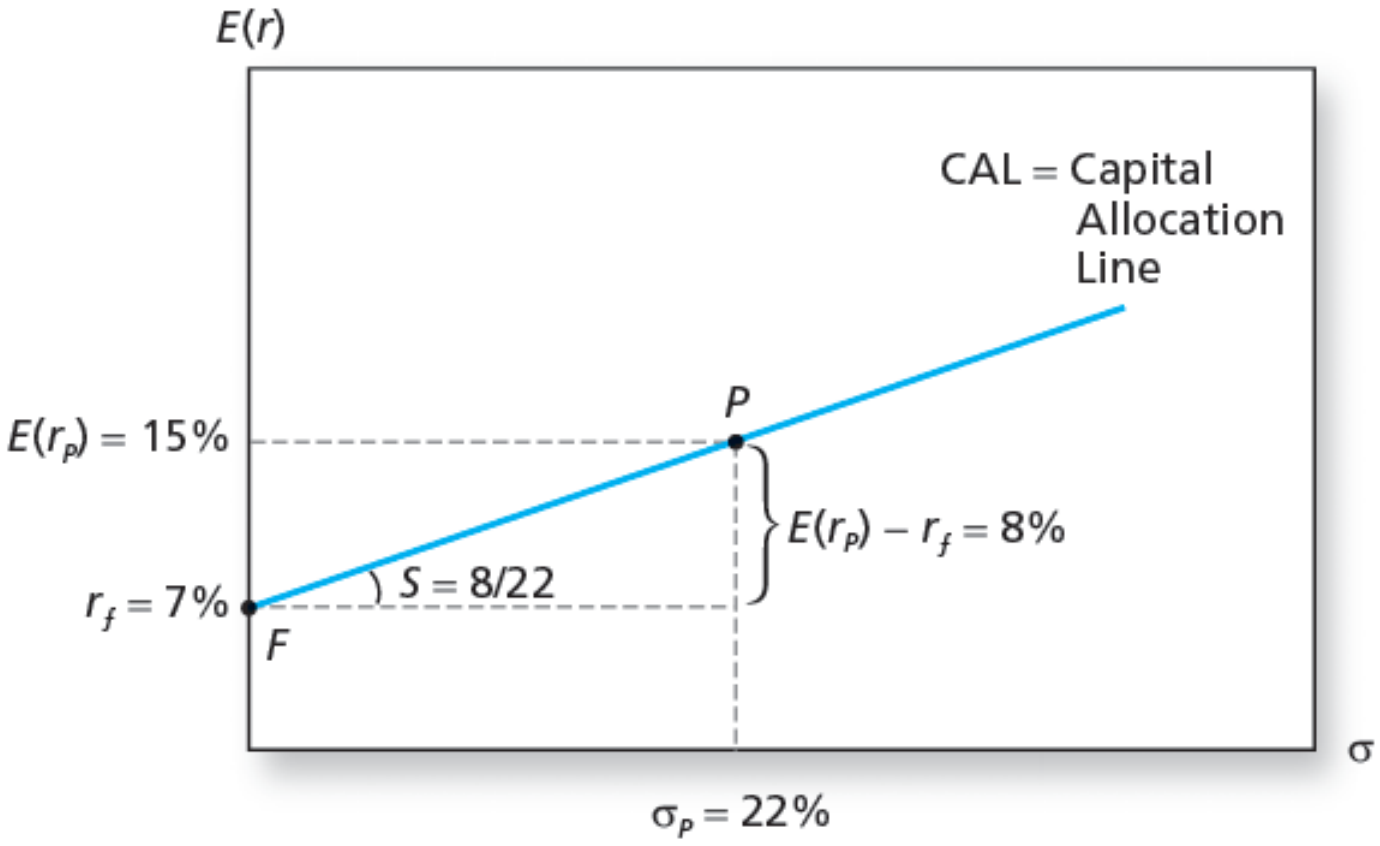
The Capital Allocation Line

CAL depicts the relation between return r_C and standard deviation σ_C: σ_C is x-axis and $E(r_C)$ is y-axis

- Write $E(r_C)$ as a function of σ_C
- Know that $E(r_C) = r_f + y(E(r_p) - r_f)$
- Sub. $y = \sigma_C/\sigma_P$: $E(r_C) = r_f + \sigma_C/\sigma_P(E(r_p) - r_f)$
- The slope of CML is $1/\sigma_P(E(r_p) - r_f)$
- The intercept of CML is r_f

CAL

- Slope is $\frac{E(r_P)-r_f}{\sigma_P}$, which is
  - 8%/22%=8/22
- Intercept is 7%



Levered Position

- Suppose the investment budget is $300,000 and our investor borrows an additional $120,000, investing the total available funds in the risky asset.
- Return on risky portfolio is 15%. If investors can borrow at the (risk-free) rate of r_f = 7%.
- This is a levered position in the risky asset, financed in part by borrowing.

Levered Position

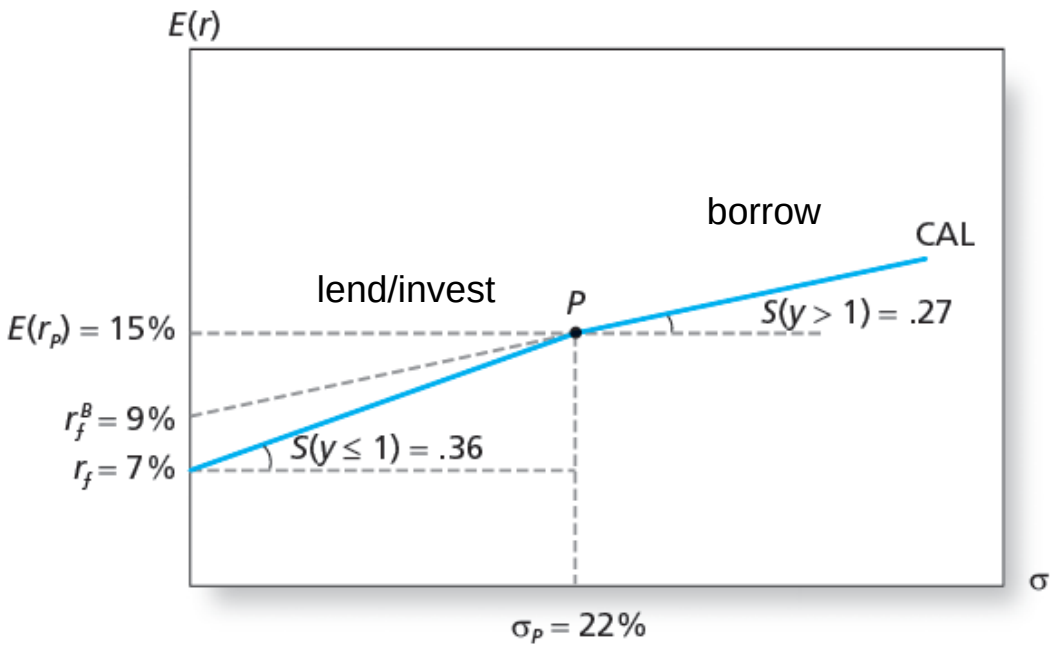
- y= 420,000/ 300,000 = 1.4
- $E(r_C) = y\,E(r_p) + (1 - y)r_f$
- or
- $E(r_C) = r_f + y\,(E(r_p) - r_f)$
- $= 18.2\%$

Borrowing rate is not lending rate

- Capital allocation line with leverage
  - Lend at r_f = 7% and borrow at r_f = 9%
    - Lending range slope = 8/22 = 0.36
    - Borrowing range slope = 6/22 = 0.27
  - CAL kinks at *P*

CAL when Borrowing rate is not lending rate

- The slope is different between $y > 1$ and $y \leq 1$



Utility Levels for Various Positions in Risky Assets

Expected return of the complete portfolio: $r_f + y\left(E\left(r_p\right) - r_f\right)$, where $r_f = 7\%, E\left(r_p\right) = 15\%$

S.D. of the complete portfolio: $y\sigma_p$, where $\sigma_p = 22\%$

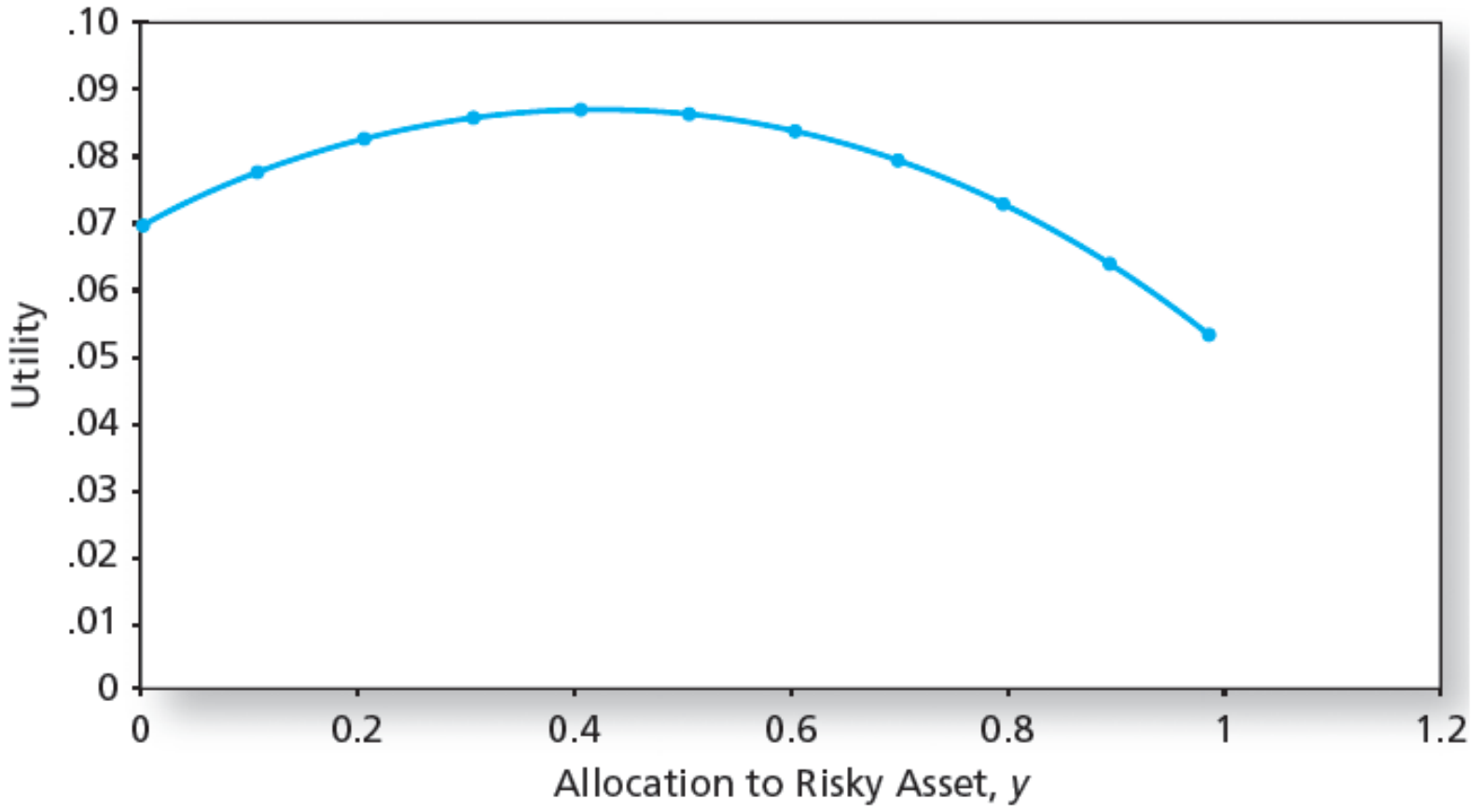
- The investor must choose one optimal portfolio *C (that is the one with the highest utility),* from the set of feasible choices.

Table 6.4

Utility levels for various positions in risky assets (y) for an investor with risk aversion $A = 4$

(1) y	(2) $E(r_C)$	(3) σ_C	(4) $U = E(r) - ½A\sigma^2$
0	.070	0	.0700
0.1	.078	.022	.0770
0.2	.086	.044	.0821
0.3	.094	.066	.0853
0.4	.102	.088	.0865
0.5	.110	.110	.0858
0.6	.118	.132	.0832
0.7	.126	.154	.0786
0.8	.134	.176	.0720
0.9	.142	.198	.0636
1.0	.150	.220	.0532

Utility as a Function of Allocation to the Risky Asset, *y*



Utility Maxmization

- $U = E(r_C) - 1/2A\sigma_C^2$
- We write the problem as follows:
- $\max_y U = \max_y \; y\mathrm{E}\left(r_p\right) + (1-y)r_\mathrm{f} - 1/2A\mathrm{y}^2\sigma_C^2$

Utility Maximization

- $\max_{y} U = \max_{y} \; y\mathrm{E}\left(r_p\right) + (1-y)r_{\mathrm{f}} - 1/2A\mathrm{y}^2\sigma_p^2$
- The maximization problem is solved by setting the derivative of this expression to zero, hence
- $y^* = \dfrac{E\left(r_p\right) - r_f}{A\sigma_p^2}$

Calculations of Indifference Curves

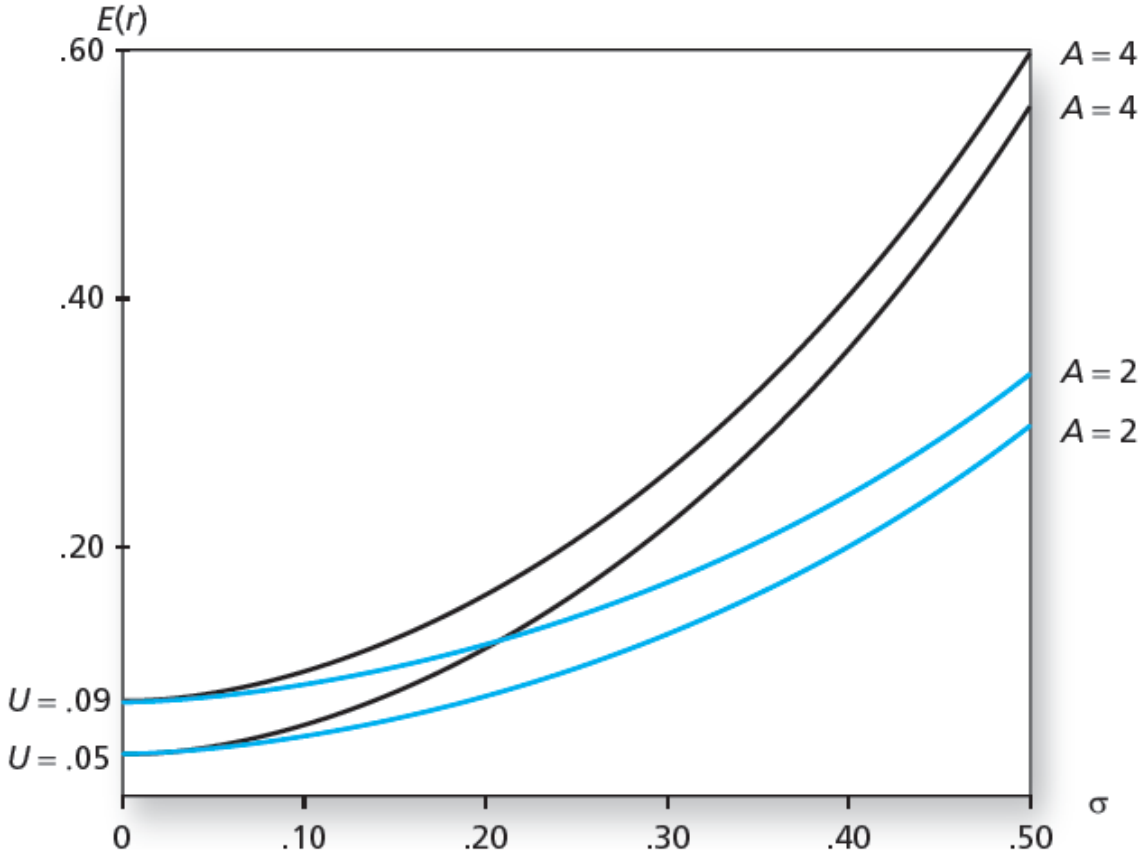
Example, when $\sigma = 0.15, A = 2$

$$E(r) = 0.05 + 0.5 \times 2 \times 0.15^2 = 0.0725$$

	$A = 2$		$A = 4$	
σ	$U = .05$	$U = .09$	$U = .05$	$U = .09$
0	.0500	.0900	.050	.090
.05	.0525	.0925	.055	.095
.10	.0600	.1000	.070	.110
.15	.0725	.1125	.095	.135
.20	.0900	.1300	.130	.170
.25	.1125	.1525	.175	.215
.30	.1400	.1800	.230	.270
.35	.1725	.2125	.295	.335
.40	.2100	.2500	.370	.410
.45	.2525	.2925	.455	.495
.50	.3000	.3400	.550	.590

Plot the relationship between E(r) and σ

- Portfolios on higher indifference curves offer a higher expected return for any given level of risk
- more risk-averse investors have steeper indifference curves than less risk-averse investor



Expected Returns on Four Indifference Curves and the CAL

σ	U = .07	U = .078	U = .08653	U = .094	CAL
0	.0700	.0780	.0865	.0940	.0700
.02	.0708	.0788	.0873	.0948	.0773
.04	.0732	.0812	.0897	.0972	.0845
.06	.0772	.0852	.0937	.1012	.0918
.08	.0828	.0908	.0993	.1068	.0991
.0902	.0863	.0943	.1028	.1103	.1028
.10	.0900	.0980	.1065	.1140	.1064
.12	.0988	.1068	.1153	.1228	.1136
.14	.1092	.1172	.1257	.1332	.1209
.18	.1348	.1428	.1513	.1588	.1355
.22	.1668	.1748	.1833	.1908	.1500
.26	.2052	.2132	.2217	.2292	.1645
.30	.2500	.2580	.2665	.2740	.1791

A=4

- Utility Function

when $\sigma_c = 0.0902, U = 0.08653,$

$$E(r) = 0.08653 + 0.5 \times 4 \times 0.0902^2$$
$$= 10.28\%$$

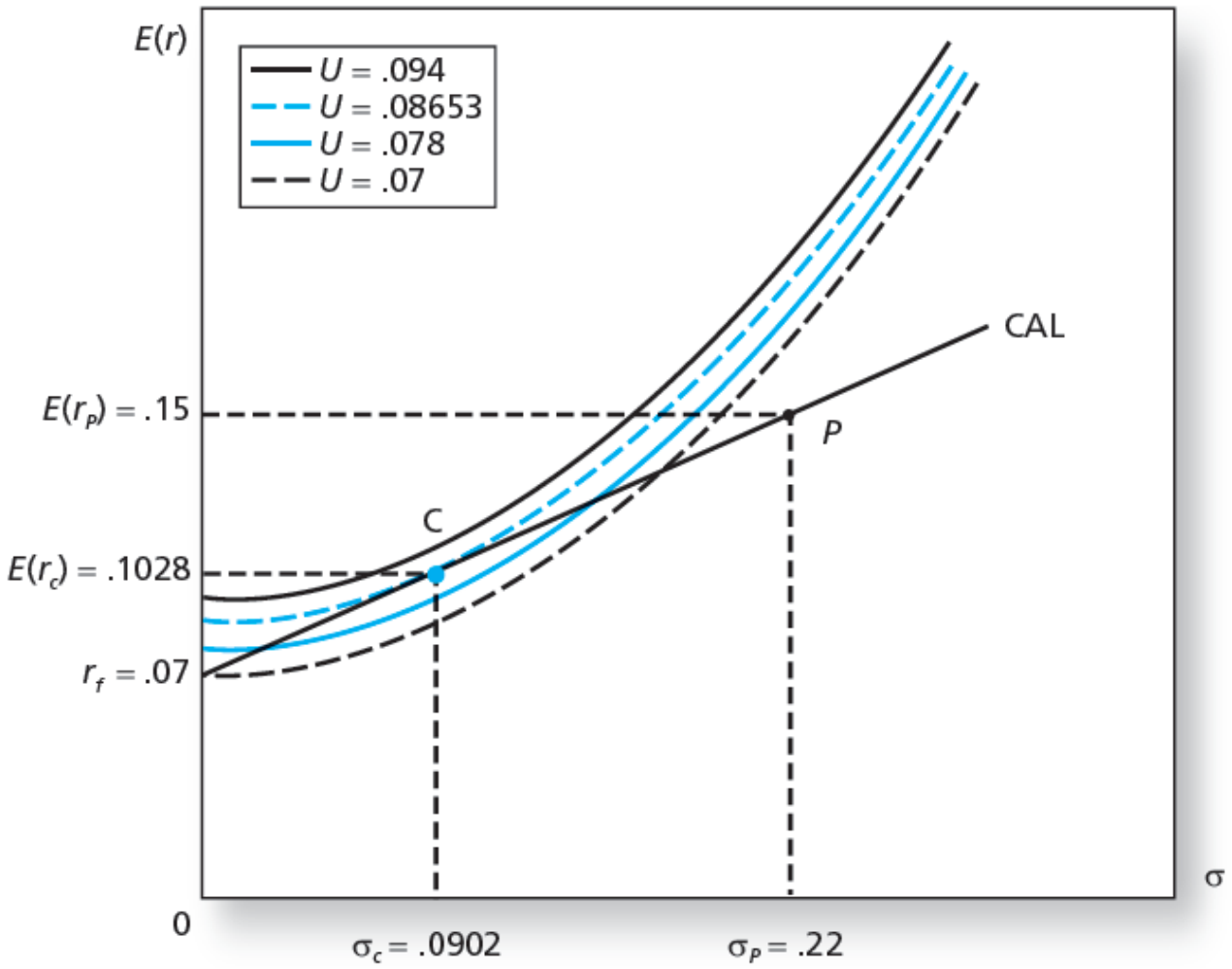
- CAL (recall that $r_f = 7\%, E(r_p) = 15\%$):

$$y = \sigma_c \backslash \sigma_p = 0.41$$
$$E(r) = r_f + y\,[E(r_p) - r_f] = 10.28\%$$

Implication:

- One can build a portfolio with the risk of 0.0902 by allocating 41% of wealth to risky portfolio. It produces the best possible return of 10.28%. Such portfolio offers you a utility score of 0.08653.
- Higher than 0.08653 is not feasible.
- Lower than 0.08653 is not ideal.

Finding the Optimal Complete Portfolio



- Identify the *highest possible* indifference curve (or utility score) that still touches the CAL.
- That curve is tangent to the CAL, and the tangency point corresponds to the standard deviation and expected return of the optimal complete portfolio.
- In this example, the optimal complete portfolio is achieved when $y = 0.41, E(r) = 10.28\%, \sigma = 0.0902$

Finding the Optimal Complete Portfolio

- Recall that $y^* = \frac{E(r_p) - r_f}{A\sigma_p^2}$
  - $y^* = \frac{15\% - 7\%}{4 \times (22\%)^2} = 0.41$
- Which is the same as that derived from the tangency points between indifference curves and CML.
- The mathematic expression for the tangency points is

$$\max_{y} \; E(r_c) - 1/2Ay^2\sigma_c^2 \quad s.t. \;\; E(r_c) = r_f + y[E(r_p) - r_f]$$

Passive Strategies: The Capital Market Line

The Capital Market Line (CML)

- Is a capital allocation line formed investment in two passive portfolios:
  1. Virtually risk-free short-term T-bills (or a money market fund)
  2. Fund of common stocks that mimics a broad market index (such as SP500 index)

Exercise

You manage a risky portfolio with an expected rate of return of 18% and a standard deviation of 28%. The T-bill rate is 8%. Consider three scenarios:

- ✓Suppose that your client decides to invest in your risky portfolio a fraction of *y* of the total investment budget so that the overall portfolio will have an expected rate of return of 16%.
- ✓Suppose that your client prefers to invest in your fund a proportion *y* that maximizes the expected return on the complete portfolio subject to the constraint that the complete portfolio's standard deviation will not exceed 18%.
- ✓Suppose Your client's degree of risk aversion is A = 3.5.

Compute the proportion y and the expected return of complete portfolio for each scenario.

Solution 1

$y \times 10\% + 8\% = 16\%$

solve for y, $y = 80\%$

$$E(r_C) = 16\%$$

Solution 2

$y \times 18\% \leq 28\%$

solve for y, $y \leq \frac{18}{28}$

$$E(r_C) \leq \frac{18}{28} \times 10\% + 8\% = 14.43\%$$

Solution 3

$$y = \frac{E(r_p) - r_f}{A\sigma^2}$$

$$y = 36.4\%$$

$$E(r_C) = 36.4\% \times 10\% + 8\% = 11.64\%$$